DSA ASSIGNMENT

1. Greedy Algorithms:

- Greedy algorithms make the most optimal choice at each step with the hope of finding the global optimum.

-Example: In the coin change problem, if you want to make change for ₹12 using ₹1, ₹5, and ₹10 coins, a greedy algorithm will first take the ₹10 coin, then the ₹1 coins two times.

2. Time Complexity, Space Complexity, Data Structures in Use:

- Greedy Algorithms: These vary by problem.
  - Example: Coin Change:
    - Time Complexity: O(n)
    - Space Complexity: O(1)
    - Data Structures: Arrays

3. Dynamic Programming (DP):

- DP is an optimization technique that solves complex problems by breaking them down into simpler subproblems and storing the results of subproblems to avoid computing the same results multiple times.

- Types:
  - Top-Down Approach (Memoization)
  - Bottom-Up Approach (Tabulation)
- Example Algorithms:
  - Fibonacci Sequence
  - Knapsack Problem
  - Longest Common Subsequence

4. Time Complexity, Space Complexity, Data Structures in Use for DP:

- Example: Fibonacci Sequence:
  - Time Complexity: O(n)
  - Space Complexity: O(n) for memoization, O(1) for tabulation

- Data Structures: Arrays, Hash Tables (for memoization)